

PWM & analog

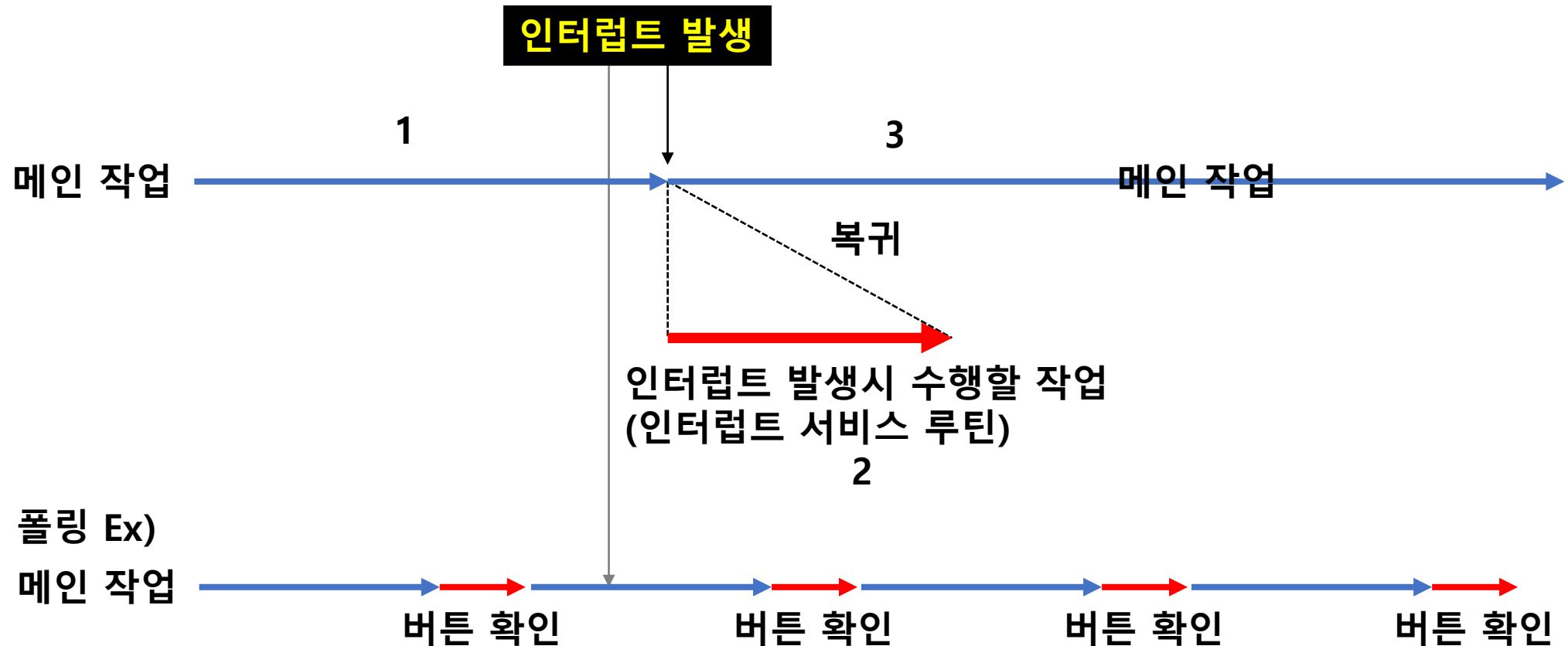
마이크로프로세서 종합 설계. 9주차.

오늘의 목표

1. 외부인터럽트의 이해 및 인터럽트 실험 (폴링 vs 인터럽트)
2. PWM의 이해와 PWM을 이용한 전력 제어 실험
3. analogRead를 이용한 강도(Power) 측정

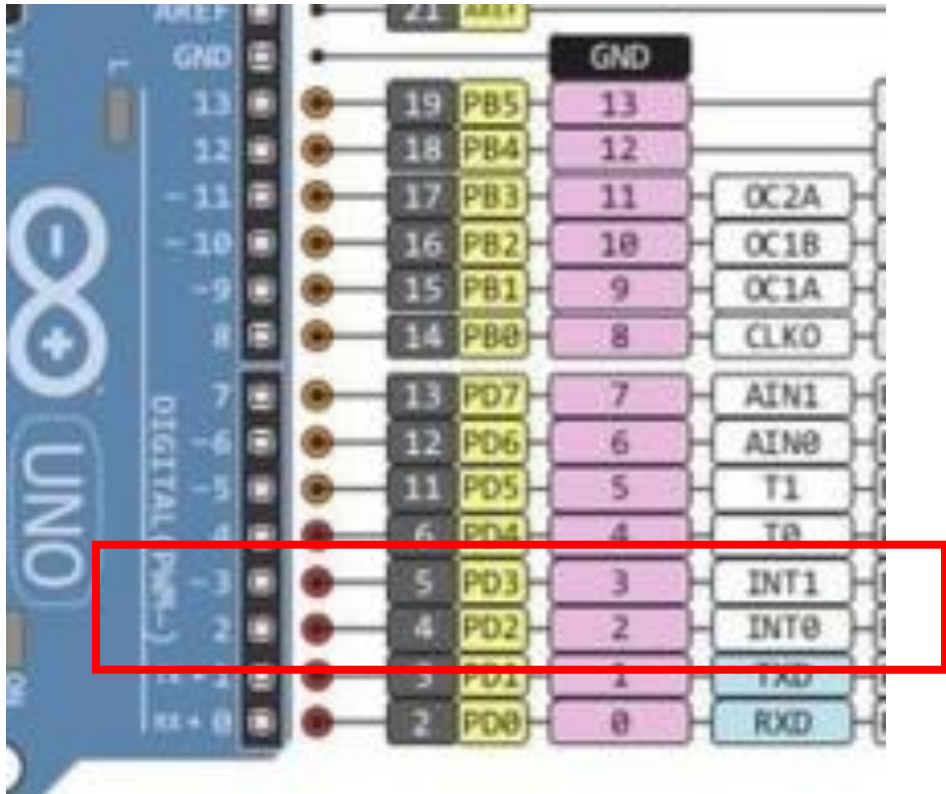
외부 인터럽트(External Interrupt)

- 폴링 vs 인터럽트



외부 인터럽트(External Interrupt)

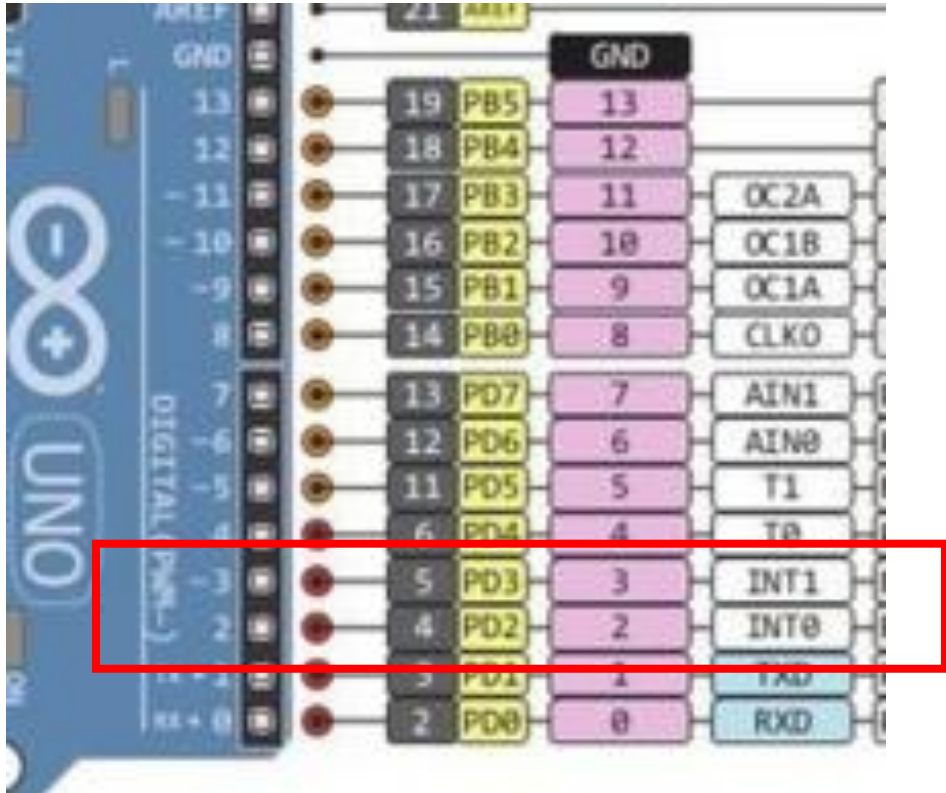
- 폴링 vs 인터럽트



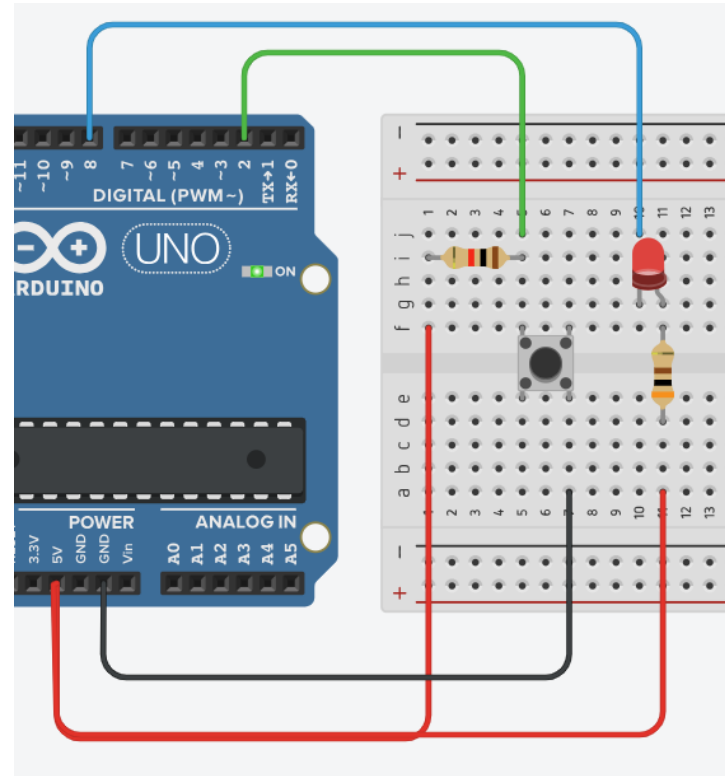
- INT1 : Interrupt #1
- INT0 : Interrupt #0

외부 인터럽트(External Interrupt)

- 플링 vs 인터럽트



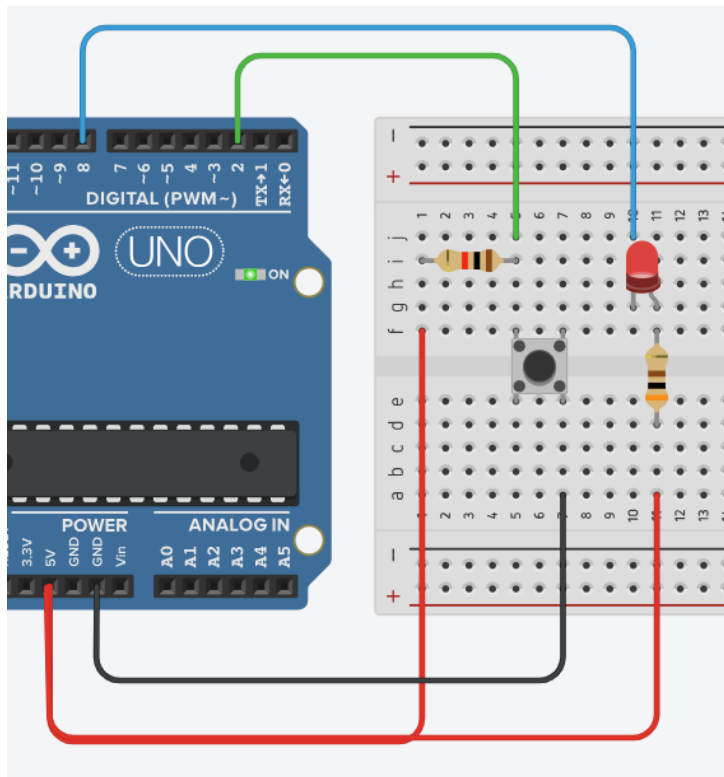
2 → INT0 : Interrupt #0



외부 인터럽트(External Interrupt)

- 폴링 vs 인터럽트

2 → INT0 : Interrupt #0



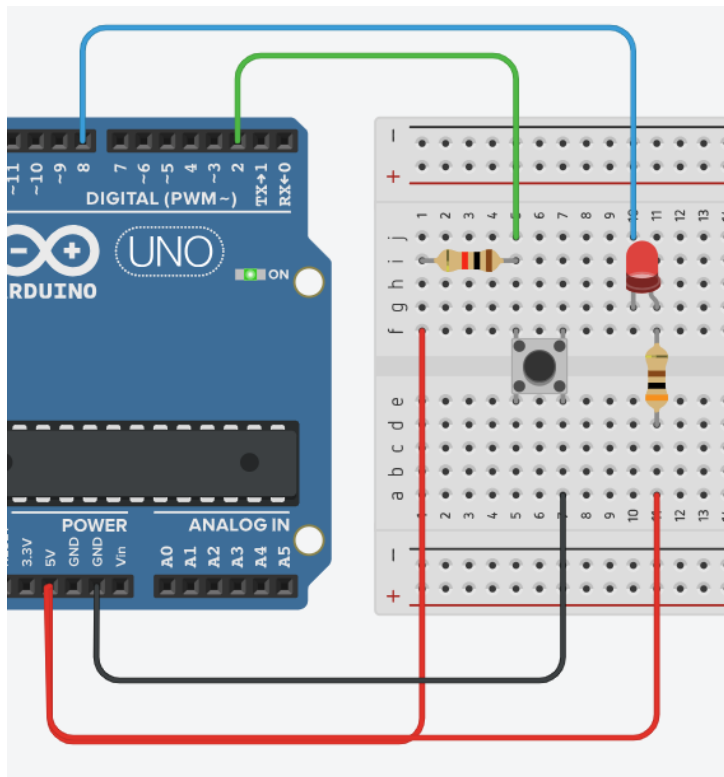
인터럽트 발동 조건 (mode)

모드	상태
LOW	핀이 LOW일때
CHANGE	LOW->HIGH or HIGH->LOW로 변할 때
RISING	LOW ->HIGH일때
FALLING	HIGH -> LOW일때
HIGH	핀이 HIGH일때

외부 인터럽트(External Interrupt)

• 폴링 vs 인터럽트

2 → INT0 : Interrupt #0



```
attachInterrupt( digitalPinToInterrupt(핀번호), 서비스루틴함수명, 모드 );
```

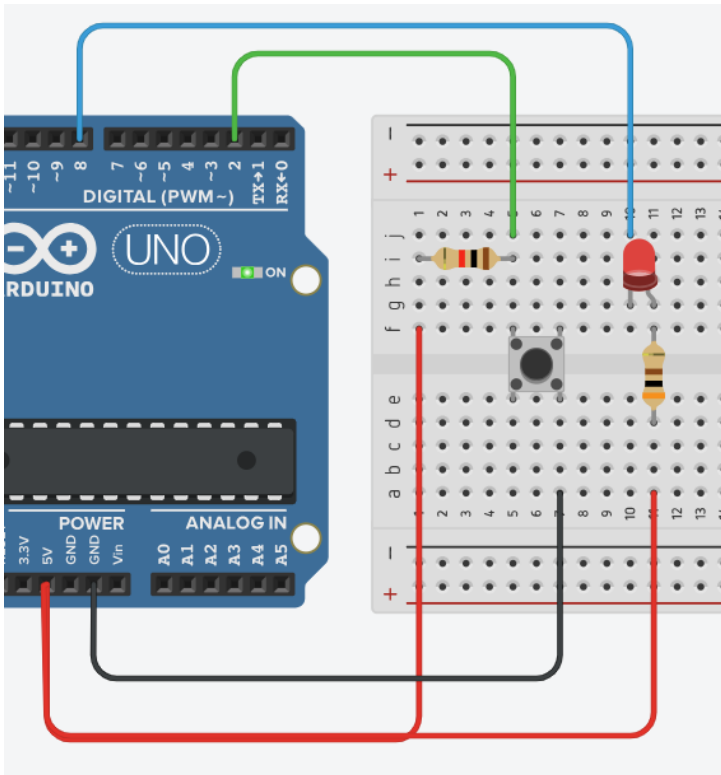
인터럽트 발동 조건 (mode)

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외부 인터럽트(External Interrupt)

• 폴링 vs 인터럽트

2 → INT0 : Interrupt #0



```
attachInterrupt( digitalPinToInterrupt(2), ExINT, FALLING );
```

```
attachInterrupt( digitalPinToInterrupt(핀번호), 서비스루틴함수명, 모드 );
```

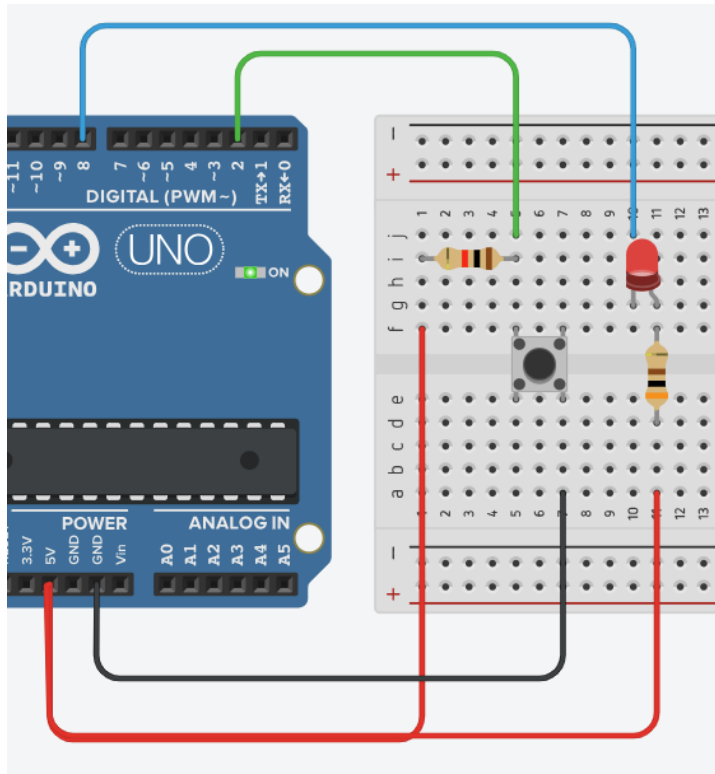
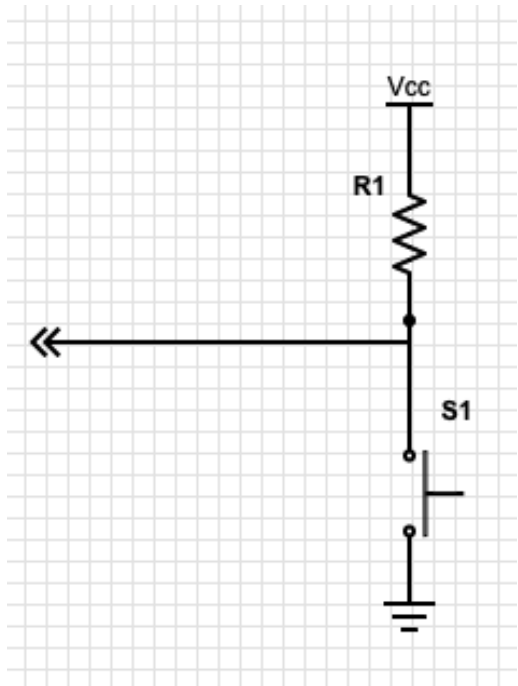
인터럽트 발동 조건 (mode)

모드	상태
LOW	핀이 LOW일때
CHANGE	LOW->HIGH or HIGH->LOW로 변할 때
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FALLING	HIGH -> LOW일때
HIGH	핀이 HIGH일때

외부 인터럽트(External Interrupt)

• 폴링 vs 인터럽트

`attachInterrupt( digitalPinToInterrupt(2), ExINT, FALLING );`



```
void setup()
{
  pinMode(8, INPUT) ;
  pinMode(2, OUTPUT) ;

  attachInterrupt( digitalPinToInterrupt(2), ExINT, FALLING );

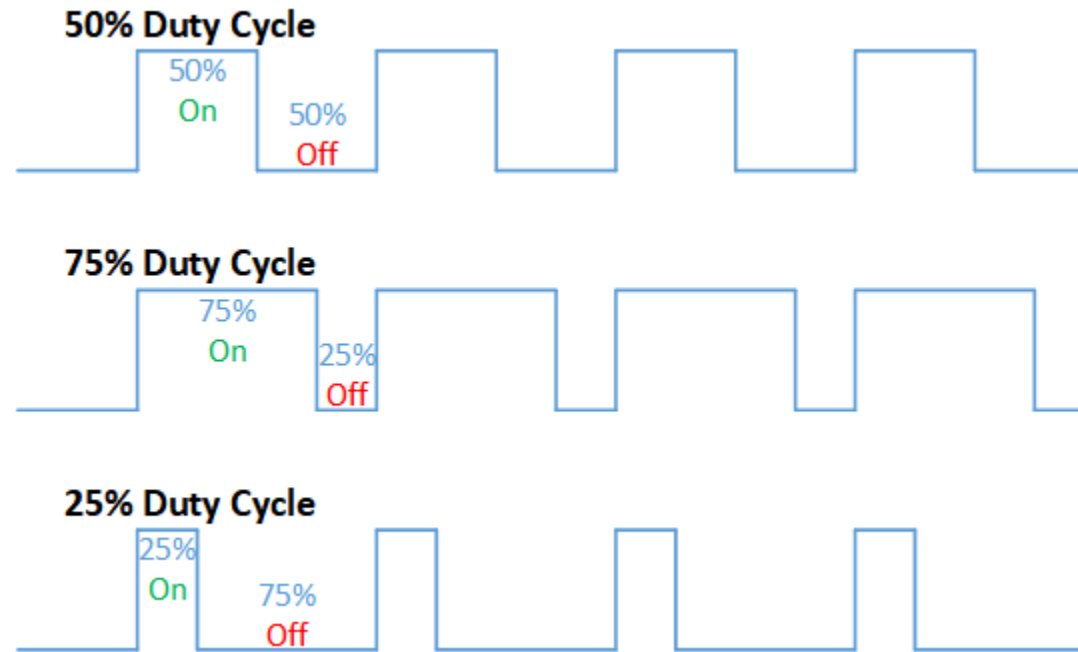
  Serial.begin(9600) ;
}

void loop()
{
  digitalWrite(8, 0) ;
  delay(1000) ;

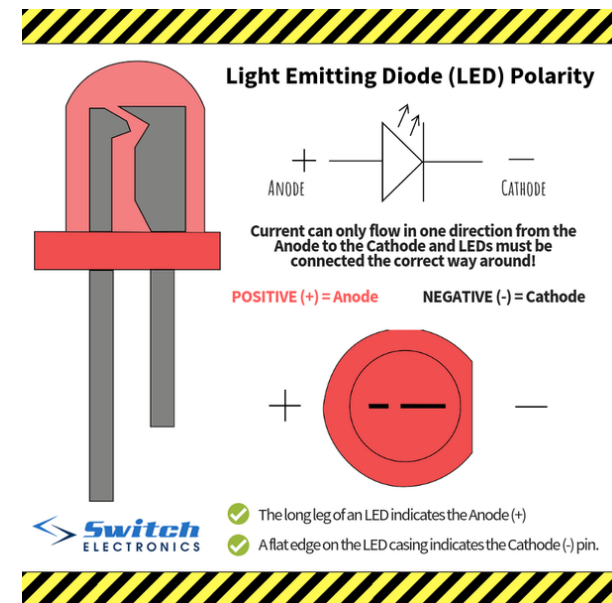
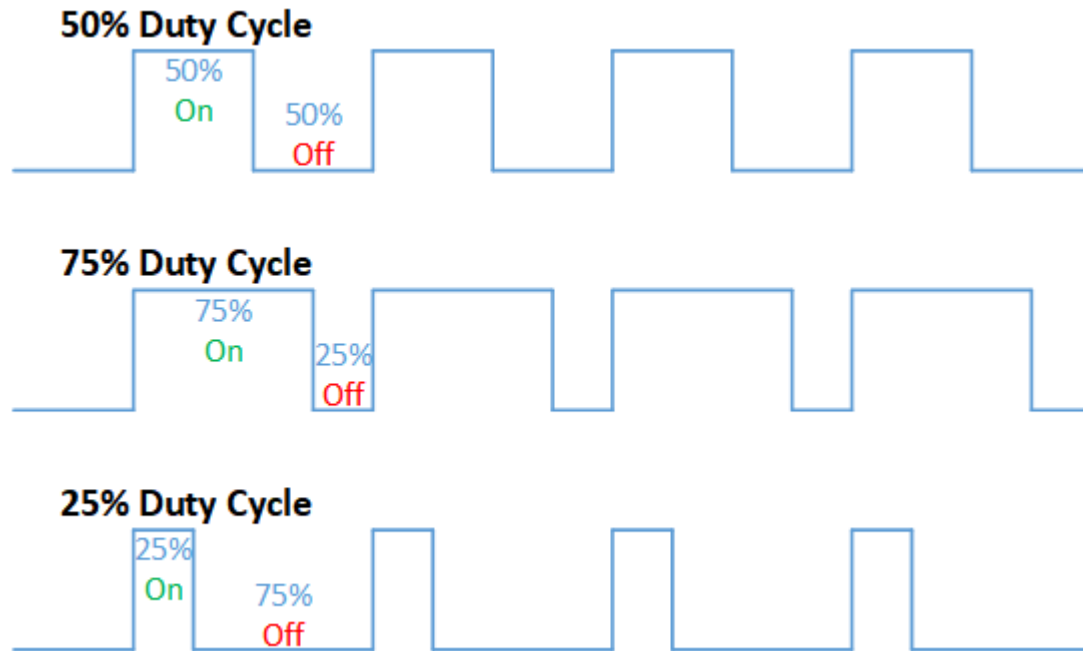
  digitalWrite(8, 1) ;
  delay(1000) ;
}

void ExINT()
{
  Serial.println("ExINT") ;
}
```

PWM(Pulse Width Modulation)

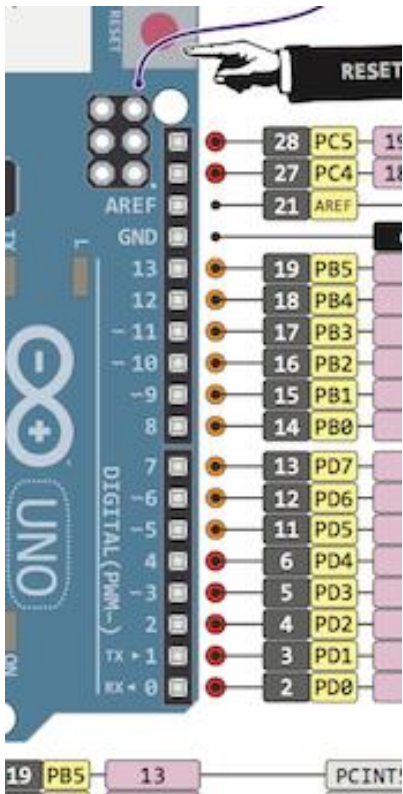


PWM을 이용한 LED 밝기 제어



디지털 입출력 관련 API

- PWM(디지털 출력) 관련 명령

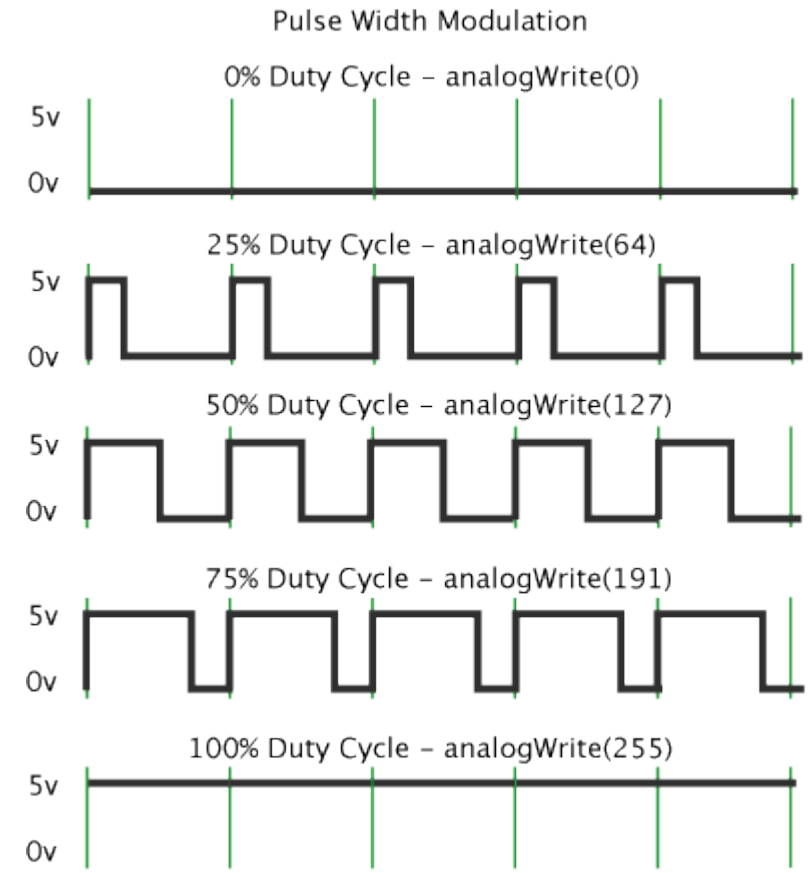
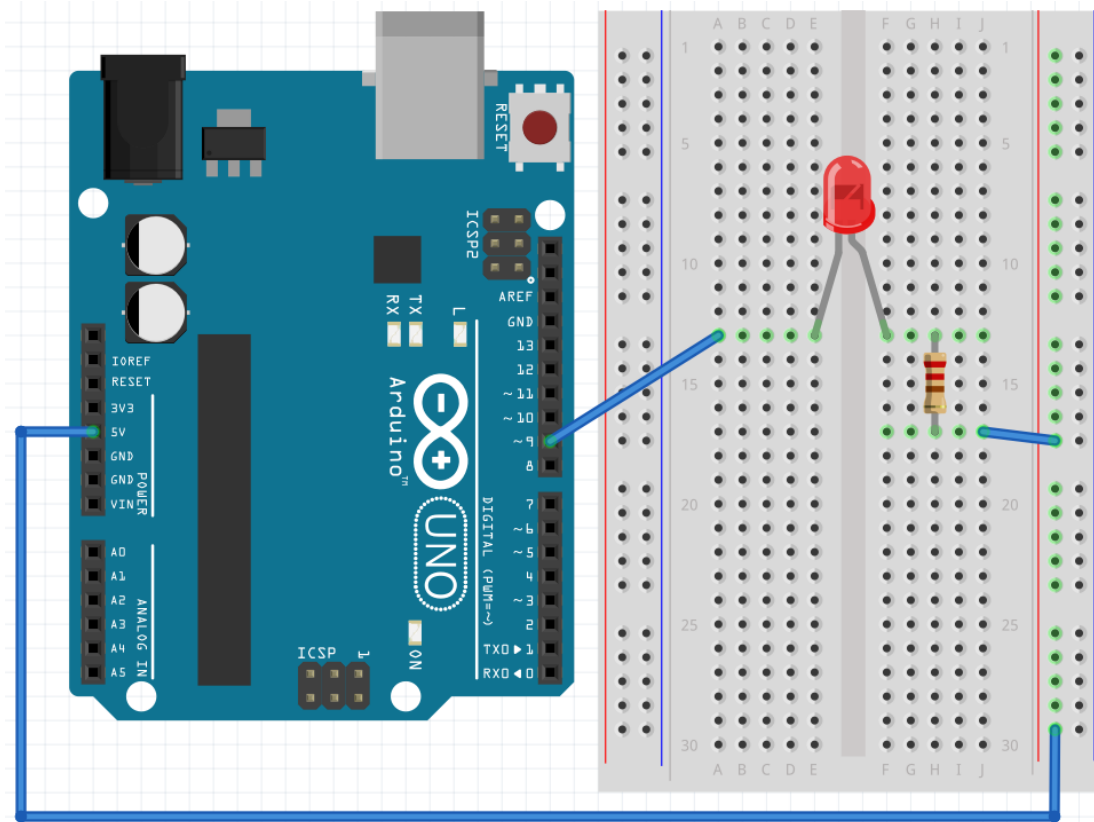


analogWrite(핀번호, Duty Cycle) ;

- `analogWrite(9, 0) ;`
- `analogWrite(9, 128) ;`
- `analogWrite(9, 255) ;`

아두이노를 이용한 LED 밝기제어 예제

- 함수 : `analogWrite(핀번호, duty cycle)`

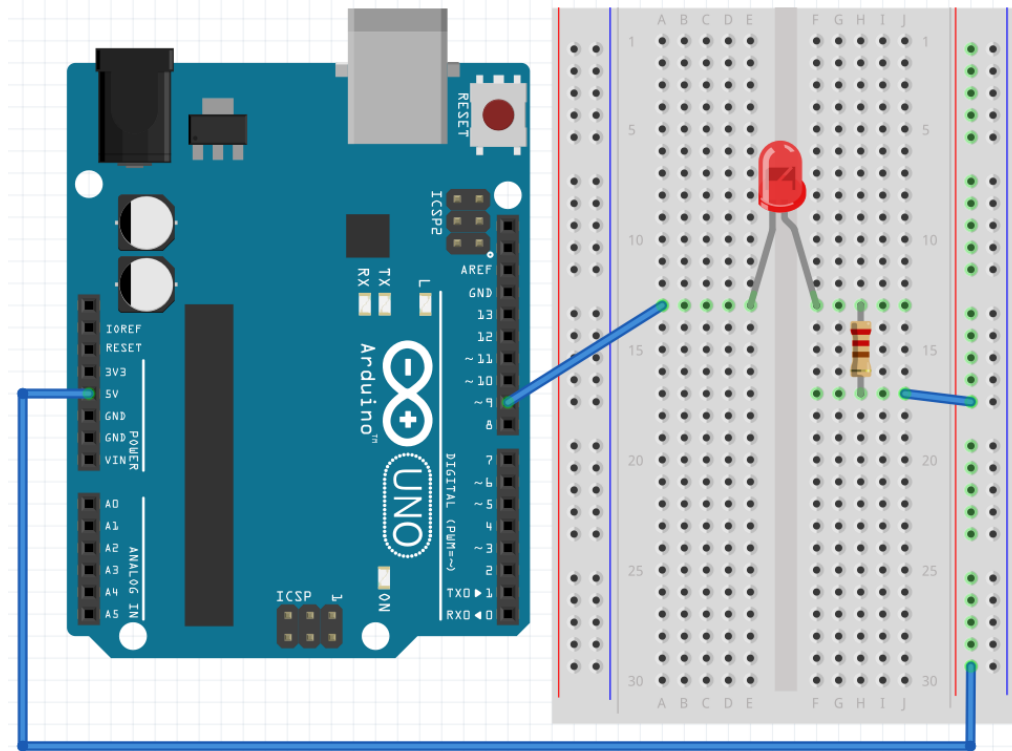


아두이노를 이용한 LED 밝기제어 예제

- 함수 : `analogWrite(핀번호, duty cycle)`

```
void setup() {  
  pinMode(9, OUTPUT); // 핀을 출력으로 설정  
}  
  
void loop() {  
  analogWrite(9, 0); //analogWrite 값은 0 부터 255까지  
}
```

```
void setup() {  
  pinMode(9, OUTPUT); // 핀을 출력으로 설정  
}  
  
void loop() {  
  analogWrite(9, 255); //analogWrite 값은 0 부터 255까지  
}
```



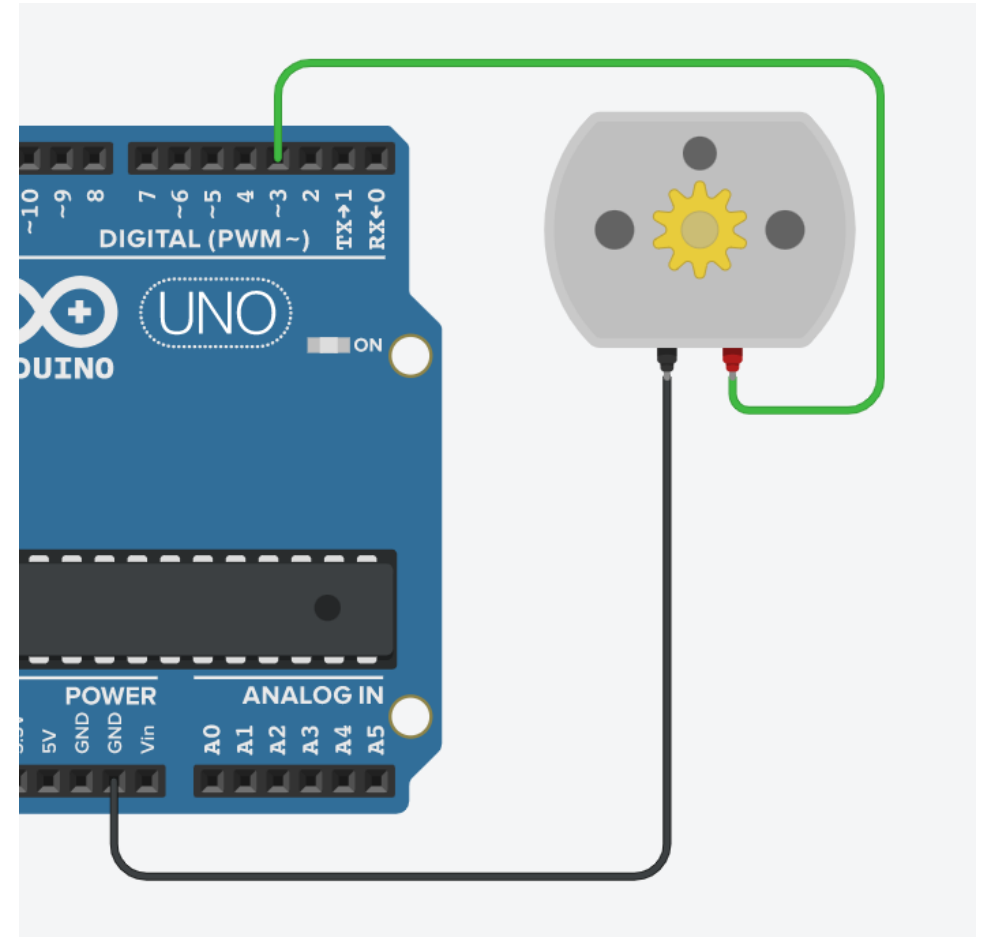
analogWrite(PWM)을 이용한 DC모터 제어

```
void setup()
{
  pinMode(3, OUTPUT); // 핀을 출력으로 설정
}

void loop()
{
  analogWrite(3, 255); //analogWrite 값은 0 부터 255까지
}
```

```
void setup()
{
  pinMode(3, OUTPUT); // 핀을 출력으로 설정
}

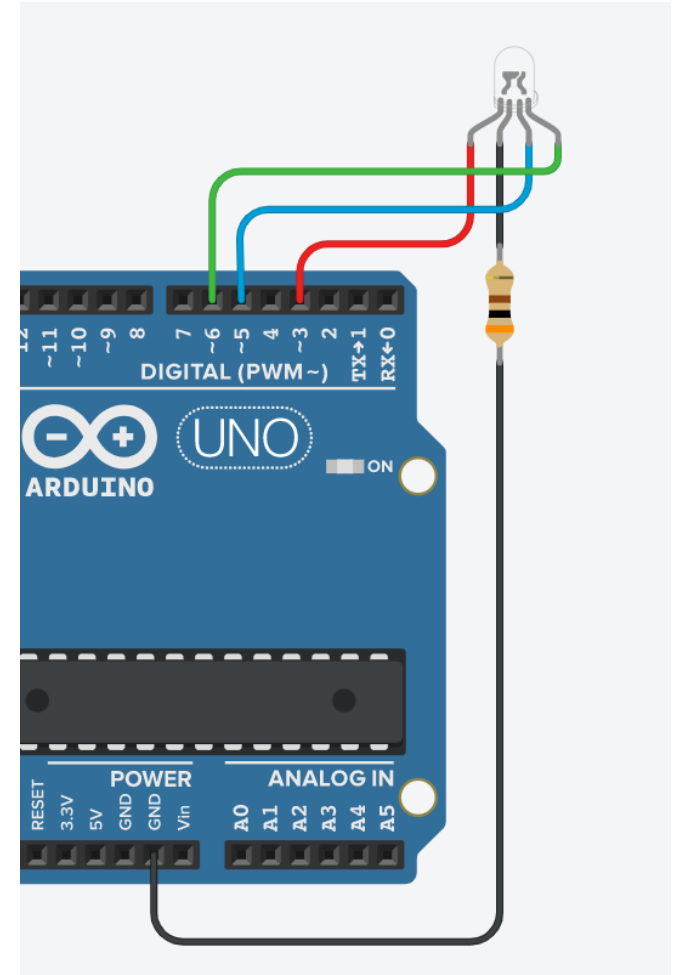
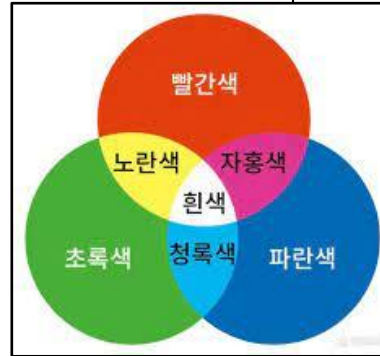
void loop()
{
  analogWrite(3, 128); //analogWrite 값은 0 부터 255까지
}
```



analogWrite(PWM)을 이용한 RGB LED제어

```
void setup()
{
  pinMode(3, OUTPUT); // 핀을 출력으로 설정
  pinMode(5, OUTPUT); // 핀을 출력으로 설정
  pinMode(6, OUTPUT); // 핀을 출력으로 설정
}
```

```
void loop()
{
  analogWrite(3, 255); //analogWrite 값은 0 부터 255까지
  analogWrite(5, 255); //analogWrite 값은 0 부터 255까지
  analogWrite(6, 0); //analogWrite 값은 0 부터 255까지
}
```



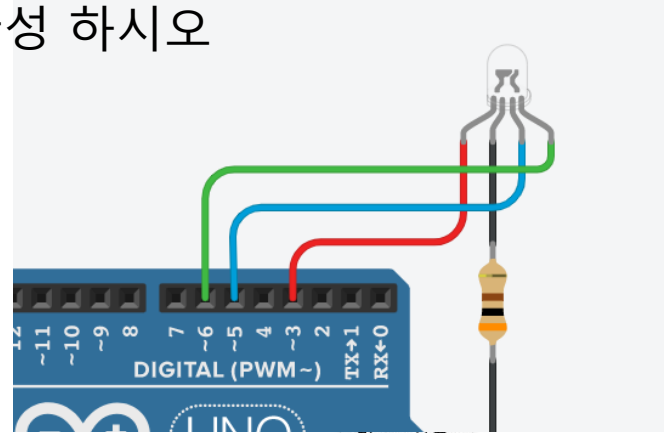
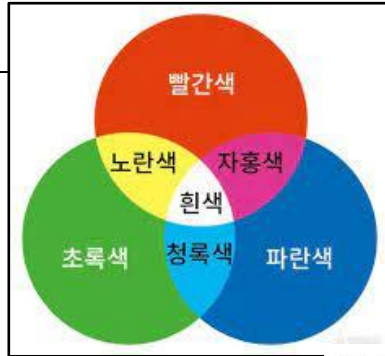
analogWrite(PWM)을 이용한 3색(full color) RGB LED제어

QUIZ : 3색 LED의 광량을 조절하여 아래의 **청록색**이 표현 되도록 코드를 완성 하시오

RGB(121, 207, 203)

```
void setup()
{
  pinMode(3, OUTPUT); // 핀을 출력으로 설정
  pinMode(5, OUTPUT); // 핀을 출력으로 설정
  pinMode(6, OUTPUT); // 핀을 출력으로 설정
}

void loop()
{
  analogWrite(3, [red]); //analogWrite 값은 0 부터 255까지
  analogWrite(5, [blue]); //analogWrite 값은 0 부터 255까지
  analogWrite(6, [green]); //analogWrite 값은 0 부터 255까지
}
```

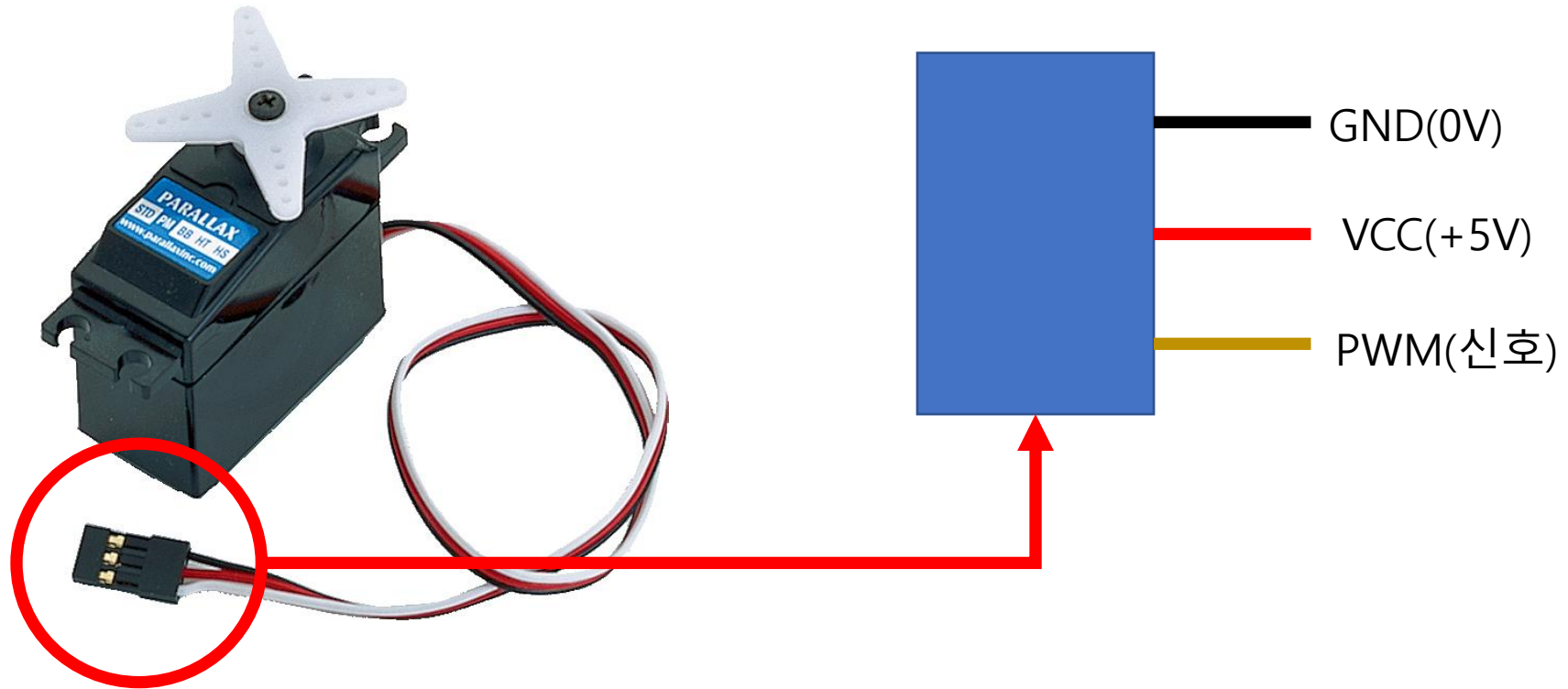


색 편집

값 81(청록)

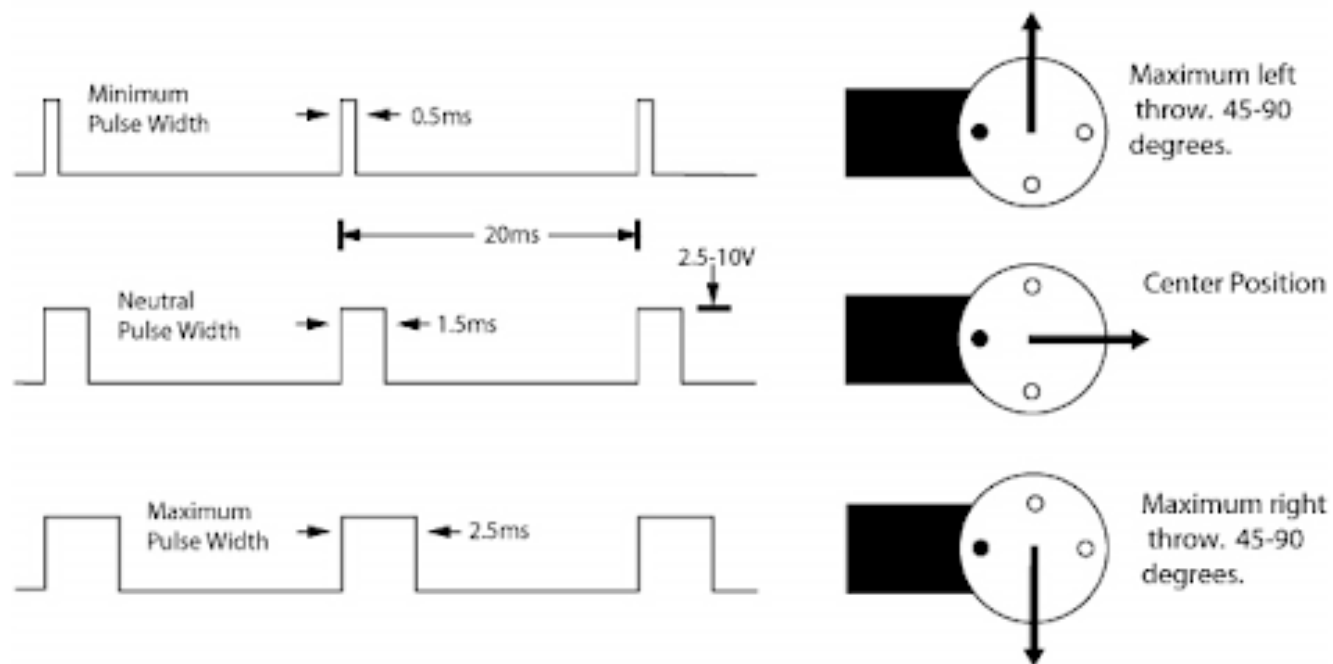


RC 서보모터



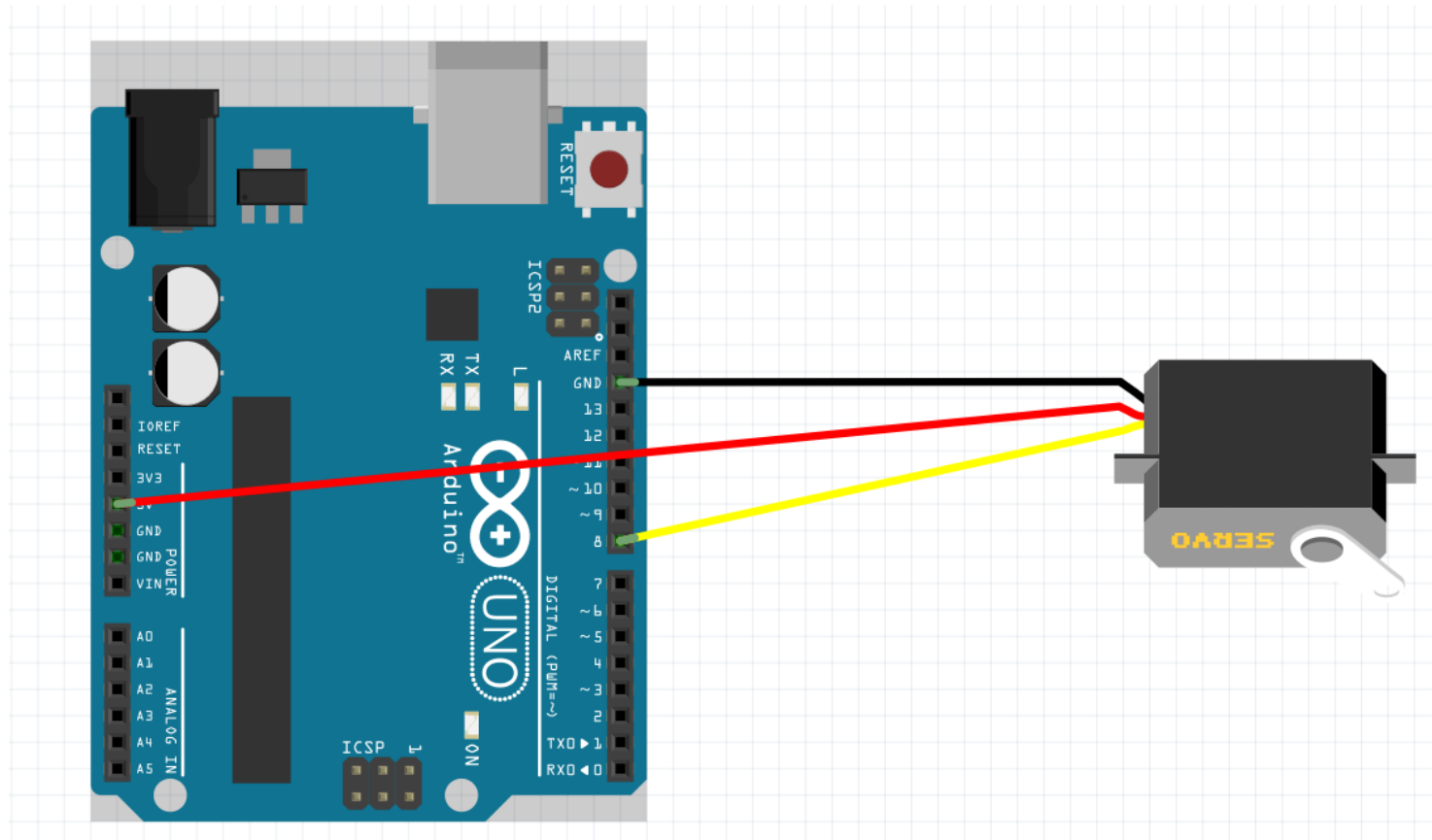
PWM을 이용한 RC 서보모터 제어

R/C Control Signal Theory



아두이노를 이용한 서보모터 제어

- 테스트 회로 구성



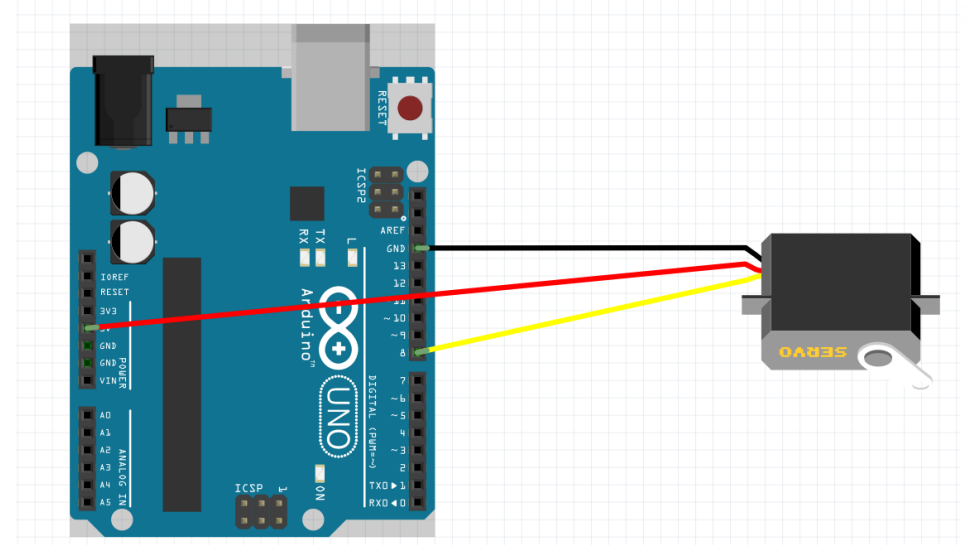
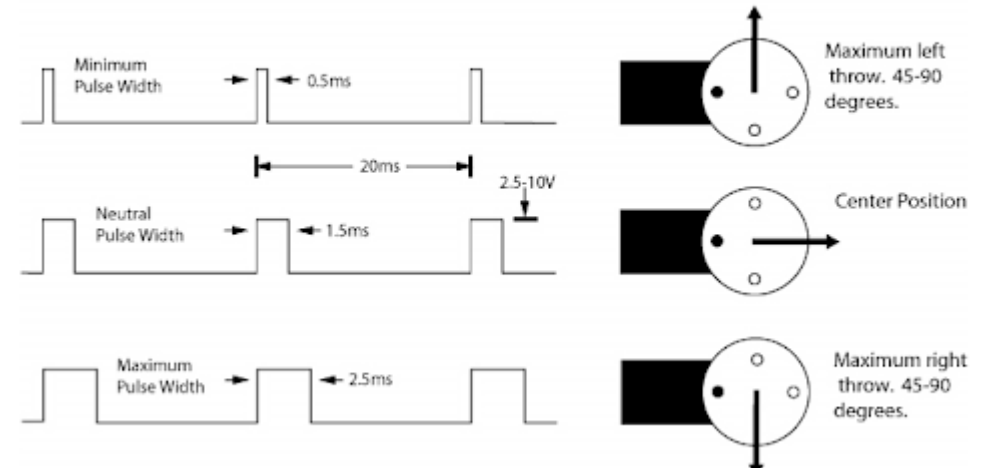
아두이노를 이용한 서보모터 제어

```
void setup()
{
  pinMode(8, OUTPUT) ;
}

void loop()
{
  digitalWrite(8, HIGH);
  delayMicroseconds(1400);

  digitalWrite(8, LOW);
  delayMicroseconds(20000-1400);
}
```

R/C Control Signal Theory



아두이노를 이용한 서보모터 제어

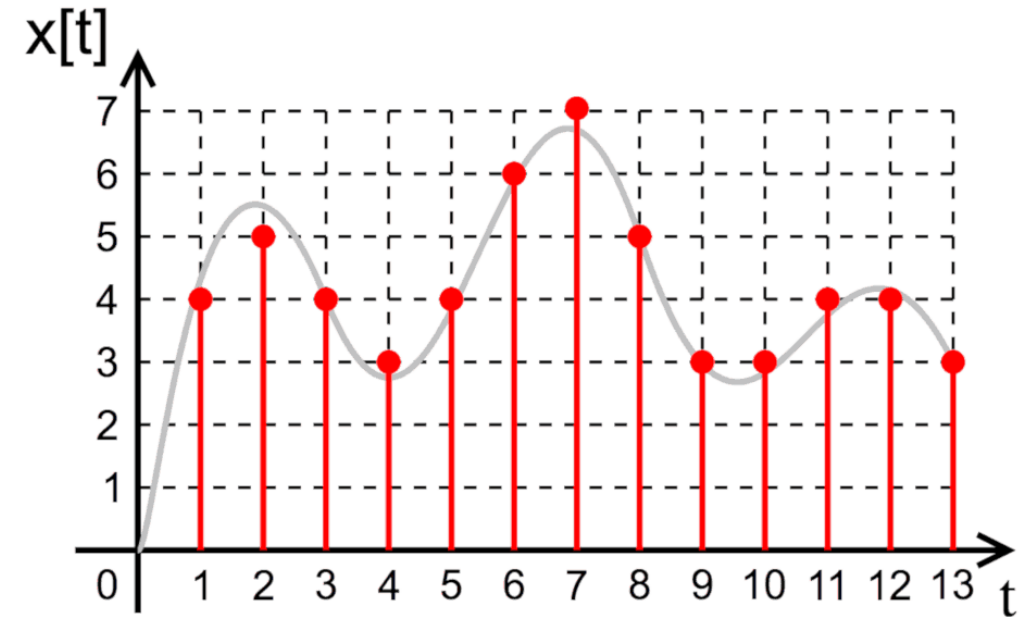
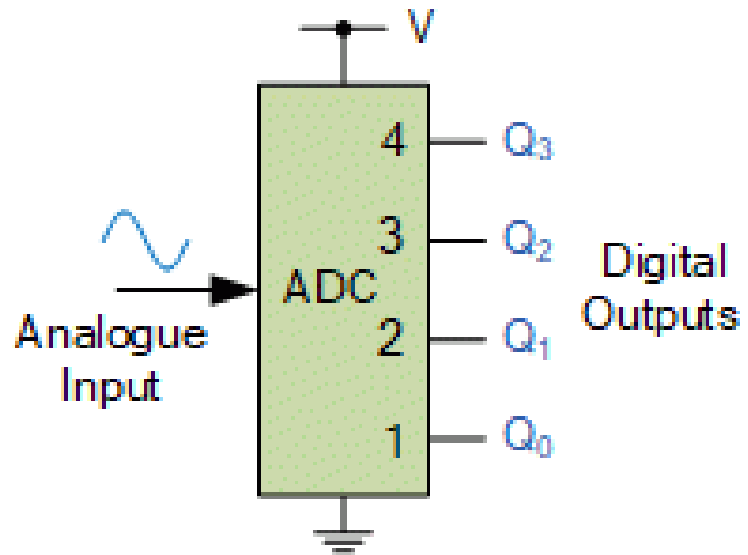
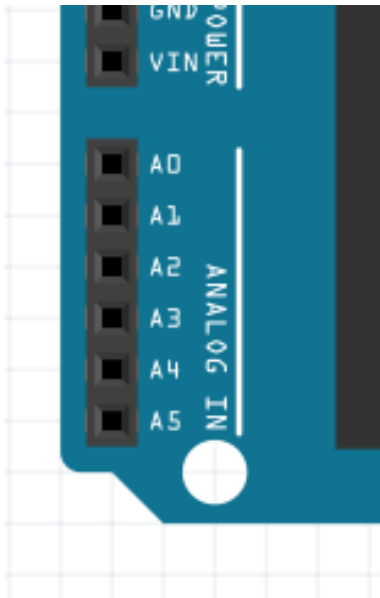
- 퀴즈

- 버튼을 누르는 동안에만 모터가 0도에 위치하고 그렇지 않으면 90도 (반대로) 이동시키는 코드를 작성하시오

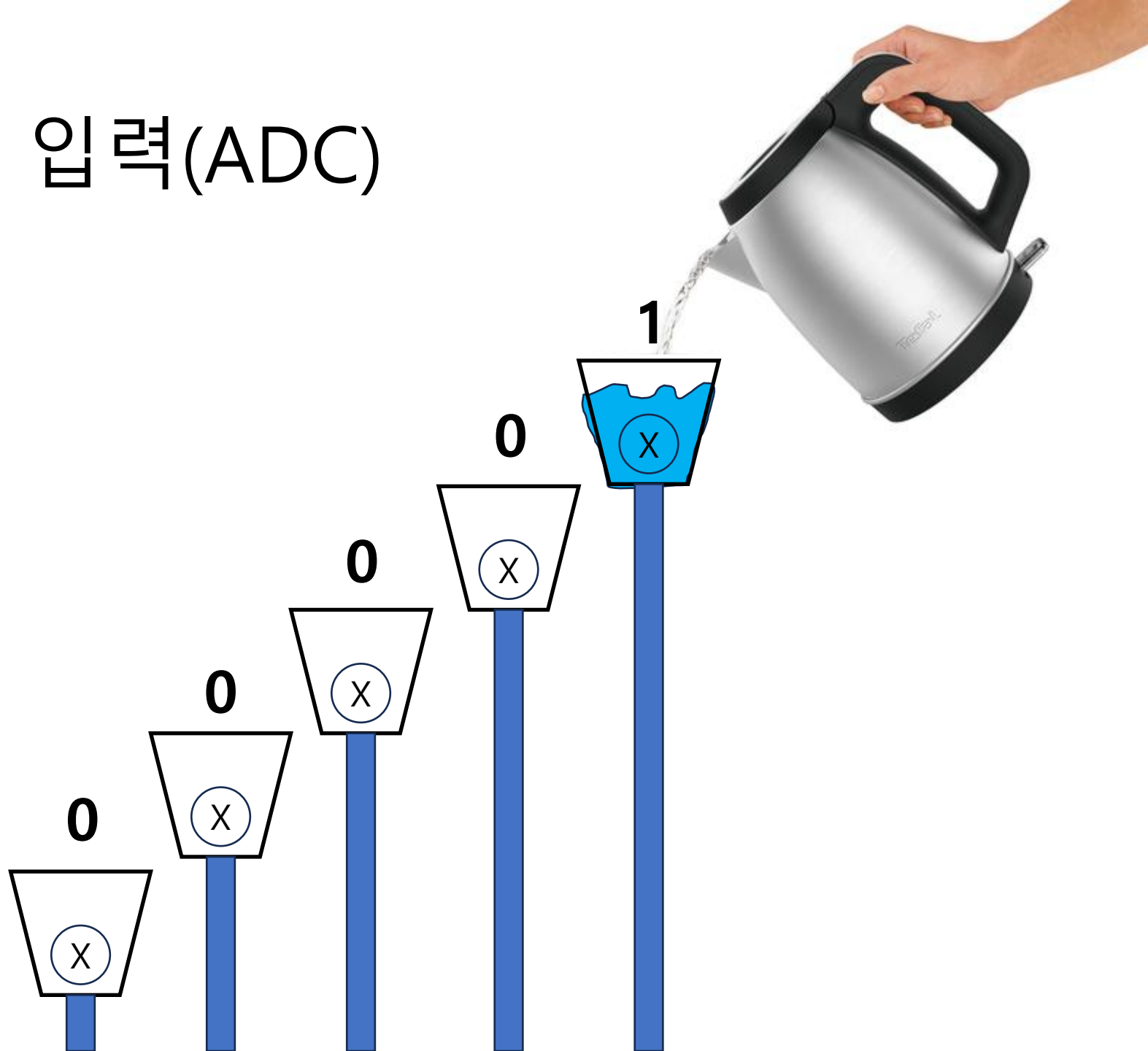
- Hint

- pinMode(핀번호, INPUT_PULLUP);
- if (digitalRead(핀번호) == LOW){} ~ else{}

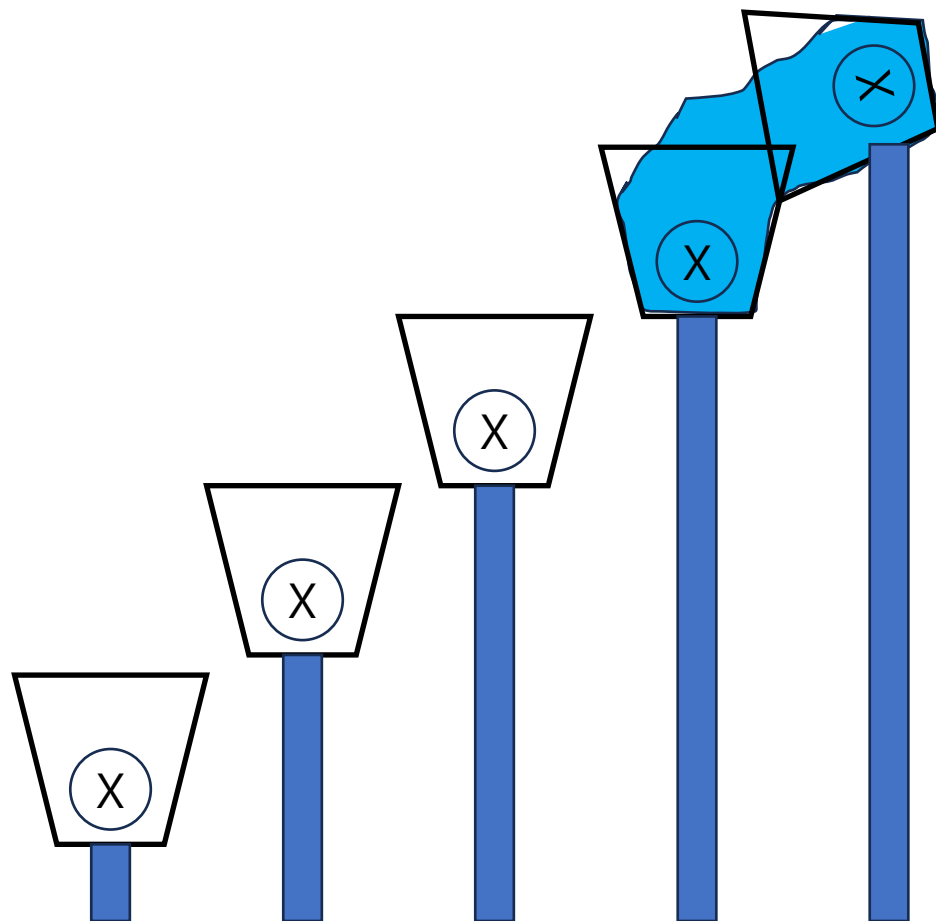
아날로그 입력(ADC)



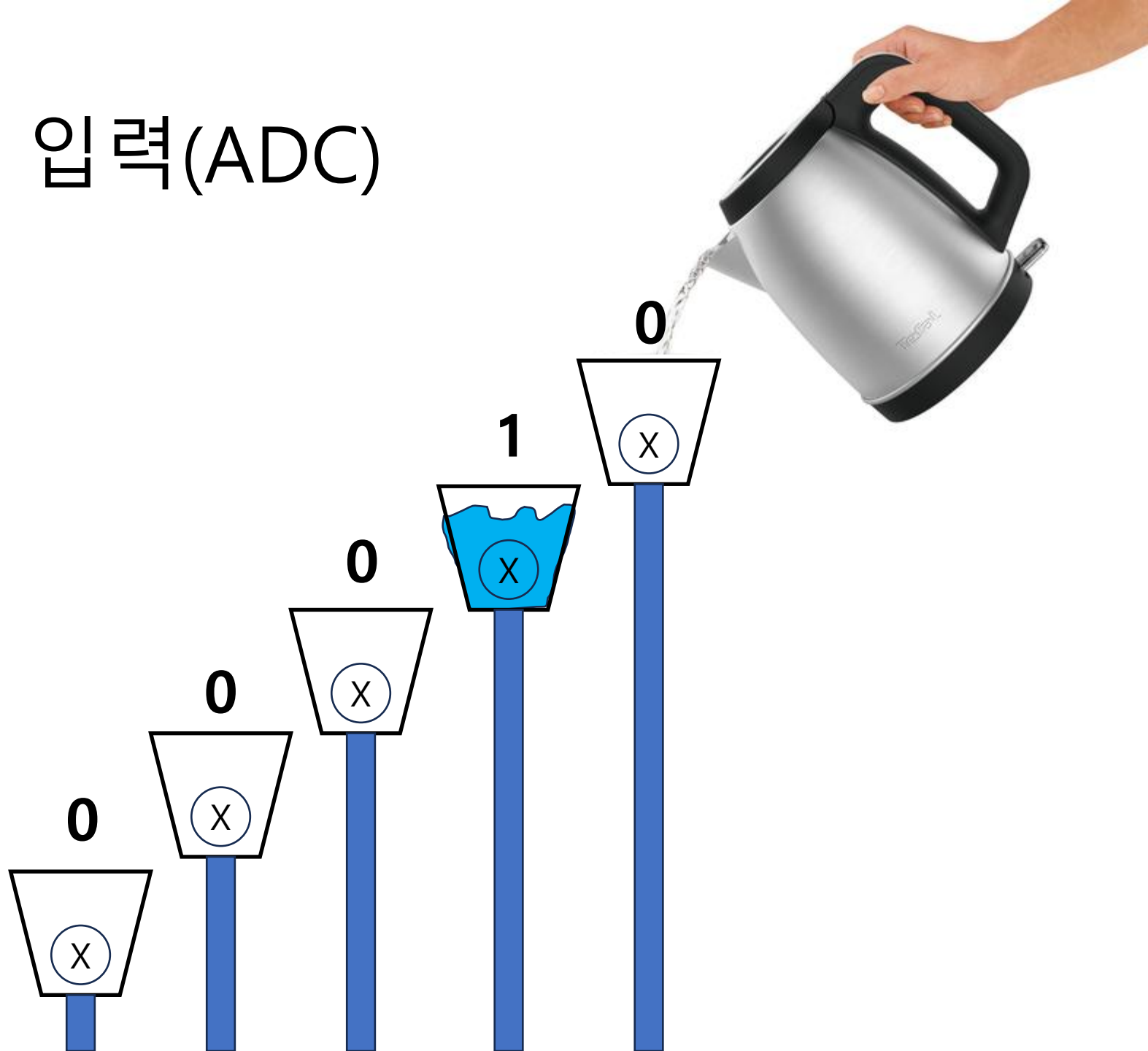
아날로그 입력(ADC)



아날로그 입력(ADC)

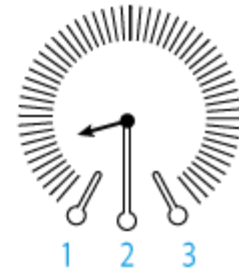
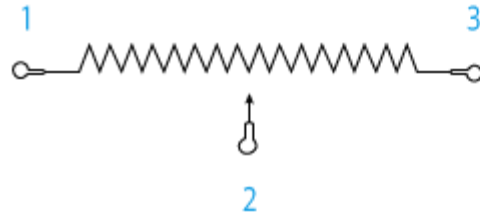


아날로그 입력(ADC)



가변저항(Potentiometer, 볼륨)

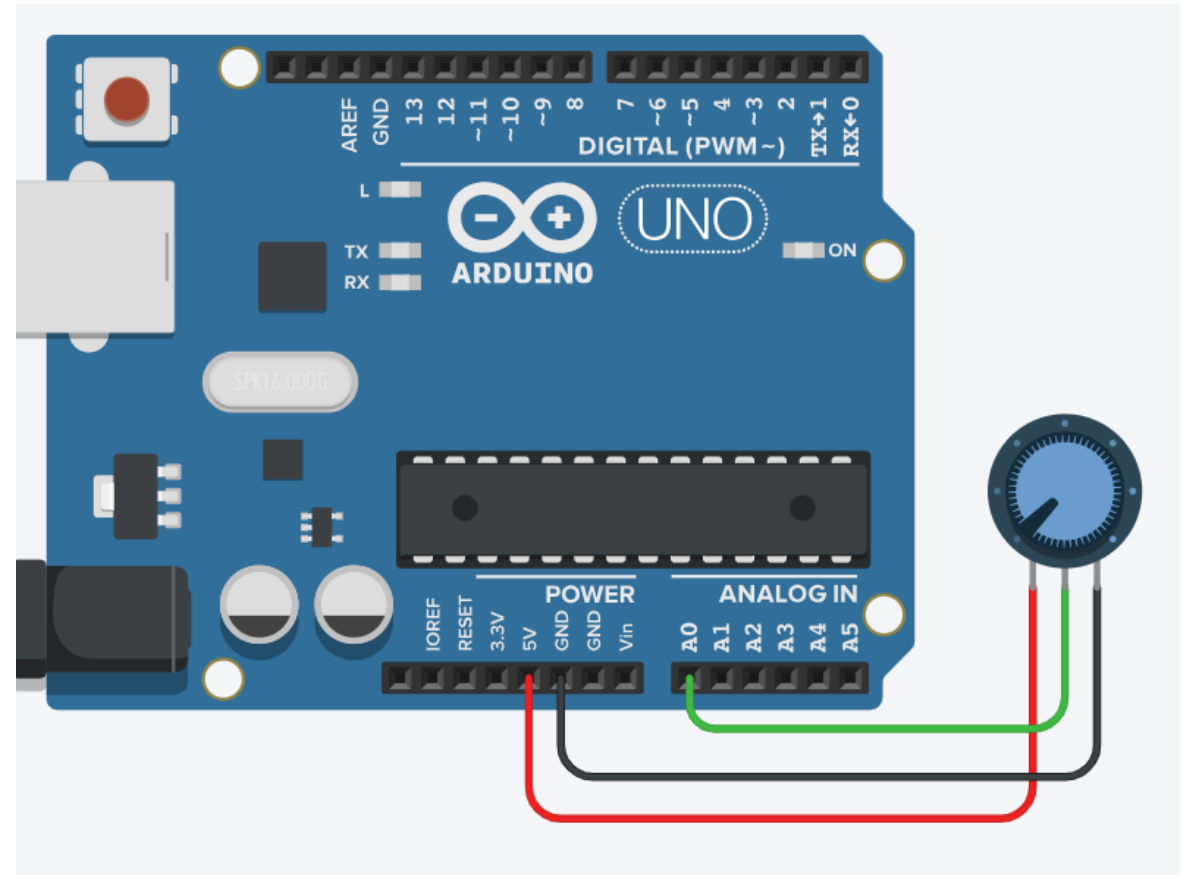
- 저항값을 변경



가변저항의 동작방식

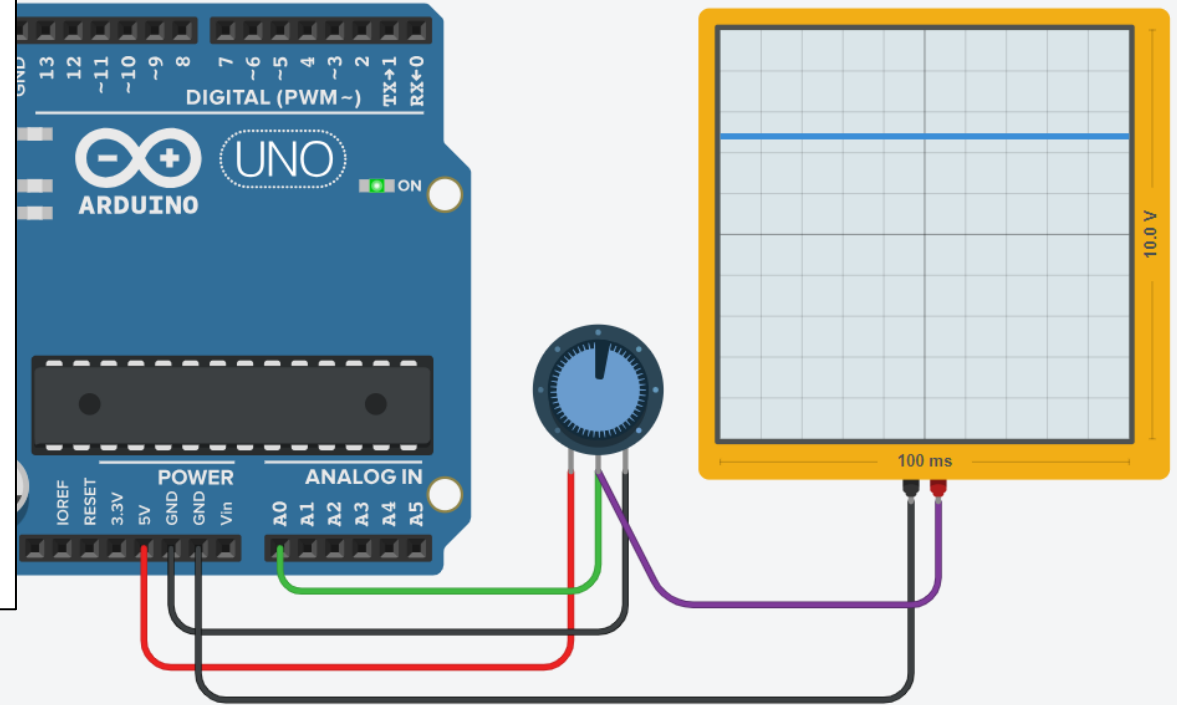
가변저항(Potentiometer, 볼륨)

```
void setup ()  
{  
  Serial.begin(9600);  
}  
  
void loop()  
{  
  int val = analogRead(A0);  
  Serial.print("Analog : ");  
  Serial.println(val);  
}
```



가변저항(Potentiometer, 볼륨)

```
void setup ()  
{  
  Serial.begin(9600);  
}  
  
void loop()  
{  
  int val = analogRead(A0);  
  Serial.print("Analog : ");  
  Serial.println(val);  
}
```



analogWrite / analogRead 실험

```
void setup ()
{
  Serial.begin(9600);
  pinMode(3, OUTPUT);
}

void loop()
{
  int val = analogRead(A0);
  Serial.print("Analog : ");
  Serial.println(val);

  val = val / 4 ;

  analogWrite(3, val);
}
```

